Technical Performance & Stability Roadmap (Android 2D-in-3D System)

System Context

- Architecture: 3D Shell (renderer/compositor) + 2D Lobby (UI/orchestrator) + Child Apps + Android OS + Play Store updates.
- Key challenge: contention between real-time rendering and lifecycle/orchestration logic.
- Primary risk: coupling between shell, lobby, and update pipeline.

Critical Execution Paths (Golden Paths)

- Cold Boot → Shell init → Lobby first frame → User interaction ready.
- Lobby → App launch → Foreground window ready → Input routed.
- App → Teardown → Shell regain focus → Lobby rehydration.

Instrumentation & Telemetry Layer

- Latency metrics: TTF (time-to-first-frame), launch duration, return latency.
- Frame metrics: long frames (>16.6ms), jank %, compositor stalls.
- Thread metrics: main-thread blocking time, binder wait time.
- Resource metrics: CPU/GPU utilization, memory growth, GC frequency, texture uploads.
- Reliability metrics: crash rate, ANR rate, launch failure codes, crash-loop detection.

Profiling Toolchain

- Android: Perfetto/System Trace, Logcat, Tombstones, Memory Profiler.
- UI: FrameMetrics/JankStats, Choreographer callbacks.
- 3D Layer: GPU profiler, render pass analysis, Unity/Native frame markers.
- Lifecycle: ActivityManager, PackageManager call tracing.

Systemic Bottleneck Categories

- Main-thread saturation: synchronous IO, PackageManager queries, asset decoding.
- IPC/binder contention: excessive cross-process calls during transitions.
- GPU stalls: overdraw, large UI surface textures, expensive shaders.
- Memory pressure: heap fragmentation, texture leaks, background process churn.
- Update instability: version skew, incompatible app updates, partial installs.

High-ROI Architectural Interventions

- Process isolation: separate Shell3D and Lobby2D processes.
- Transactional launch pipeline: preflight → launch → readiness signal → timeout/fallback.
- Crash-loop quarantine: exponential backoff + unhealthy app marking.
- Update gating: version compatibility matrix + staged rollout policies.

Optimization Strategy (Priority Order)

- Tier 1: remove main-thread blockers and lifecycle races.
- Tier 2: reduce IPC chatter and surface invalidation frequency.
- Tier 3: optimize rendering pipeline and asset loading.
- Tier 4: memory lifecycle tuning and GPU micro-optimizations.

Validation & Regression Control

- Before/after benchmarking of golden paths.
- Performance budgets enforced in CI/CD.
- Telemetry-based regression alerts.
- Cohort-based rollout with rollback triggers.

Target Technical KPIs

- Home TTF < 1s (P50), < 2s (P90).
- App launch success rate > 99.9%.
- Jank rate < 5% during transitions.
- Crash/ANR rate trending toward zero.
- Stable memory footprint over long uptime.